

SubQueries

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Assignment Name: SubQueries Assignment

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Creating new database and setting up given tables :

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department_id VARCHAR(5),  
    salary INT  
);
```

```
INSERT INTO Employee  
(emp_id, name, department_id, salary)  
VALUES  
(101,'Abhishek','D01',62000),  
(102,'Shubham','D01',58000),  
(103,'Priya','D02',67000),  
(104,'Rohit','D02',64000),  
(105,'Neha','D03',72000),  
(106,'Arman','D03',55000),  
(107,'Ravi','D04',60000),
```

```
(108,'Sneha','D04',75000),  
(109,'Kiran','D05',70000),  
(110,'Tanuja','D05',65000);
```

```
CREATE TABLE Department (  
    department_id VARCHAR(5) PRIMARY KEY,  
    department_name VARCHAR(50),  
    location VARCHAR(50)  
);
```

```
INSERT INTO Department  
(department_id, department_name, location)  
VALUES  
( 'D01','Sales','Mumbai'),  
( 'D02','Marketing','Delhi'),  
( 'D03','Finance','Pune'),  
( 'D04','HR','Bengaluru'),  
( 'D05','IT','Hyderabad');
```

```
CREATE TABLE Sales (  
    sale_id INT PRIMARY KEY,  
    emp_id INT,  
    sale_amount INT,  
    sale_date DATE,  
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id)  
);
```

```
INSERT INTO Sales  
(sale_id, emp_id, sale_amount, sale_date)  
VALUES  
(201,101,4500,'2025-01-05'),  
(202,102,7800,'2025-01-10'),  
(203,103,6700,'2025-01-14'),  
(204,104,12000,'2025-01-20'),  
(205,105,9800,'2025-02-02'),  
(206,106,10500,'2025-02-05'),  
(207,107,3200,'2025-02-09'),  
(208,108,5100,'2025-02-15'),  
(209,109,3900,'2025-02-20'),  
(210,110,7200,'2025-03-01');
```

Basic Level :

Q1. Retrieve the names of employees who earn more than the average salary of all employees.

Answer:

```
SELECT name
FROM Employee
WHERE salary >
(
    SELECT AVG(salary)
    FROM Employee
);
```

```
67 • SELECT name
68 FROM Employee
69 WHERE salary >
70 (
71     SELECT AVG(salary)
72     FROM Employee
73 );
74
```

Result Grid	Filter Rows:
name	
Priya	
Neha	
Sneha	
Kiran	
Tanuja	

Q2. Find the employees who belong to the department with the highest average salary.

Answer:

```
SELECT name
FROM Employee
WHERE department_id =
(
    SELECT department_id
    FROM Employee
    GROUP BY department_id
    ORDER BY AVG(salary) DESC
    LIMIT 1
);
```

```
75 • SELECT name
76 FROM Employee
77 WHERE department_id =
78 (
79     SELECT department_id
80     FROM Employee
81     GROUP BY department_id
82     ORDER BY AVG(salary) DESC
83     LIMIT 1
84 );
85
```

Result Grid	Filter Rows:
name	
Ravi	
Sneha	

Q3. List all employees who have made at least one sale.

Answer:

```
SELECT DISTINCT e.emp_id,
                e.name
FROM Employee e
JOIN Sales s
ON e.emp_id = s.emp_id;
```

```
86 • SELECT DISTINCT e.emp_id,
87     e.name
88 FROM Employee e
89 JOIN Sales s
90 ON e.emp_id = s.emp_id;
91
```

Result Grid				Filter Rows:
	emp_id	name		
▶	101	Abhishek		
	102	Shubham		
	103	Priya		
	104	Rohit		
	105	Neha		
	106	Arman		
	107	Ravi		
	108	Sneha		
	109	Kiran		
	110	Tanuja		

Q4. Find the employee with the highest sale amount.

Answer:

```
SELECT e.emp_id,  
       e.name,  
       s.sale_amount  
FROM Employee e  
JOIN Sales s  
ON e.emp_id = s.emp_id  
WHERE s.sale_amount =  
(  
    SELECT MAX(sale_amount)  
    FROM Sales  
);
```

```
92 • SELECT e.emp_id,  
93       e.name,  
94       s.sale_amount  
95 FROM Employee e  
96 JOIN Sales s  
97 ON e.emp_id = s.emp_id  
98 WHERE s.sale_amount =  
99 (  
100    SELECT MAX(sale_amount)  
101    FROM Sales  
102 );  
103
```

emp_id	name	sale_amount
104	Rohit	12000

Q5. Retrieve the names of employees whose salaries are higher than Shubham's salary.

Answer:

```
SELECT name  
FROM Employee  
WHERE salary >  
(  
    SELECT salary  
    FROM Employee  
    WHERE name = 'Shubham'  
);
```

```
104 • SELECT name  
105 FROM Employee  
106 WHERE salary >  
107 (  
108    SELECT salary  
109    FROM Employee  
110    WHERE name = 'Shubham'  
111 );  
112
```

name
Abhishek
Priya
Rohit
Neha
Ravi
Sneha
Kiran
Tanuja

Intermediate Level :

Q1. Find employees who work in the same department as Abhishek

Answer:

```
SELECT name  
FROM Employee
```

```

WHERE department_id =
(
    SELECT department_id
    FROM Employee
    WHERE name = 'Abhishek'
)
AND name <> 'Abhishek';

```

```

113 • SELECT name
114 FROM Employee
115 WHERE department_id =
116 (
117     SELECT department_id
118     FROM Employee
119     WHERE name = 'Abhishek'
120 )
121 AND name <> 'Abhishek';
122

```

Result Grid

name
Shubham

Q2. List departments that have at least one employee earning more than ₹60,000.

Answer:

```

SELECT DISTINCT d.department_name
FROM Department d
JOIN Employee e
ON d.department_id = e.department_id
WHERE e.salary > 60000;

```

```

123 • SELECT DISTINCT d.department_name
124 FROM Department d
125 JOIN Employee e
126 ON d.department_id = e.department_id
127 WHERE e.salary > 60000;
128

```

Result Grid

department_name
Sales
Marketing
Finance
HR
IT

Q3. Find the department name of the employee who made the highest sale.

Answer:

```

SELECT d.department_name
FROM Department d
JOIN Employee e
ON d.department_id = e.department_id
JOIN Sales s
ON e.emp_id = s.emp_id
WHERE s.sale_amount =
(
    SELECT MAX(sale_amount)
    FROM Sales
);

```

```

129 • SELECT d.department_name
130 FROM Department d
131 JOIN Employee e
132 ON d.department_id = e.department_id
133 JOIN Sales s
134 ON e.emp_id = s.emp_id
135 WHERE s.sale_amount =
136 (
137     SELECT MAX(sale_amount)
138     FROM Sales
139 );
140

```

Result Grid

department_name
Marketing

Q4. Retrieve employees who have made sales greater than the average sale amount.

Answer:

```
SELECT e.name,
       s.sale_amount
FROM Employee e
JOIN Sales s
ON e.emp_id = s.emp_id
WHERE s.sale_amount >
(
    SELECT AVG(sale_amount)
    FROM Sales
);
```

141 • SELECT e.name,
142 s.sale_amount
143 FROM Employee e
144 JOIN Sales s
145 ON e.emp_id = s.emp_id
146 WHERE s.sale_amount >
147 (
148 SELECT AVG(sale_amount)
149 FROM Sales
150);
151

Result Grid | Filter Rows: | Export:

	name	sale_amount
▶	Shubham	7800
	Rohit	12000
	Neha	9800
	Arman	10500
	Tanuja	7200

Q5. Find the total sales made by employees who earn more than the average salary.

Answer:

```
SELECT SUM(s.sale_amount) AS total_sales
FROM Sales s
JOIN Employee e
ON s.emp_id = e.emp_id
WHERE e.salary >
(
    SELECT AVG(salary)
    FROM Employee
);
```

152 • SELECT SUM(s.sale_amount) AS total_sales
153 FROM Sales s
154 JOIN Employee e
155 ON s.emp_id = e.emp_id
156 WHERE e.salary >
157 (
158 SELECT AVG(salary)
159 FROM Employee
160);
161

Result Grid | Filter Rows: | Export:

	total_sales
▶	32700

Advanced Level :

Q1. Find employees who have not made any sales.

Answer:

```
SELECT e.emp_id,
       e.name
FROM Employee e
LEFT JOIN Sales s
ON e.emp_id = s.emp_id
WHERE s.emp_id IS NULL;
```

162 • SELECT e.emp_id,
163 e.name
164 FROM Employee e
165 LEFT JOIN Sales s
166 ON e.emp_id = s.emp_id
167 WHERE s.emp_id IS NULL;
168

Result Grid | Filter Rows: | Export:

	emp_id	name
--	--------	------

Q2. List departments where the average salary is above ₹55,000.

Answer:

```
SELECT d.department_name,  
       AVG(e.salary) AS average_salary  
FROM Department d  
JOIN Employee e  
ON d.department_id = e.department_id  
GROUP BY d.department_name  
HAVING AVG(e.salary) > 55000;
```

```
169 • SELECT d.department_name,  
170        AVG(e.salary) AS average_salary  
171 FROM Department d  
172 JOIN Employee e  
173 ON d.department_id = e.department_id  
174 GROUP BY d.department_name  
175 HAVING AVG(e.salary) > 55000;  
176
```

Result Grid	Filter Rows:	Export:
department_name	average_salary	
Sales	60000.0000	
Marketing	65500.0000	
Finance	63500.0000	
HR	67500.0000	
IT	67500.0000	

Q3. Retrieve department names where the total sales exceed ₹10,000

Answer:

```
SELECT d.department_name,  
       SUM(s.sale_amount) AS total_sales  
FROM Department d  
JOIN Employee e  
ON d.department_id = e.department_id  
JOIN Sales s  
ON e.emp_id = s.emp_id  
GROUP BY d.department_name  
HAVING SUM(s.sale_amount) > 10000;
```

```
177 • SELECT d.department_name,  
178        SUM(s.sale_amount) AS total_sales  
179 FROM Department d  
180 JOIN Employee e  
181 ON d.department_id = e.department_id  
182 JOIN Sales s  
183 ON e.emp_id = s.emp_id  
184 GROUP BY d.department_name  
185 HAVING SUM(s.sale_amount) > 10000;  
186
```

Result Grid	Filter Rows:	Export:
department_name	total_sales	
Sales	12300	
Marketing	18700	
Finance	20300	
IT	11100	

Q4. Find the employee who has made the second-highest sale.

Answer:

```
SELECT e.name,  
       s.sale_amount  
FROM Employee e  
JOIN Sales s  
ON e.emp_id = s.emp_id
```

```

WHERE s.sale_amount =
(
    SELECT MAX(sale_amount)
    FROM Sales
    WHERE sale_amount <
    (
        SELECT MAX(sale_amount)
        FROM Sales
    )
);

```

```

187 • SELECT e.name,
188       s.sale_amount
189 FROM Employee e
190 JOIN Sales s
191 ON e.emp_id = s.emp_id
192 WHERE s.sale_amount =
193 (
194     SELECT MAX(sale_amount)
195     FROM Sales
196     WHERE sale_amount <
197     (
198         SELECT MAX(sale_amount)
199         FROM Sales
200     )
201 );
202

```

name	sale_amount
Arman	10500

Q5. Retrieve the names of employees whose salary is greater than the highest sale amount recorded.

Answer:

```

SELECT name,
       salary
FROM Employee
WHERE salary >
(
    SELECT MAX(sale_amount)
    FROM Sales
);

```

```

203 • SELECT name,
204       salary
205 FROM Employee
206 WHERE salary >
207 (
208     SELECT MAX(sale_amount)
209     FROM Sales
210 );
211

```

name	salary
Abhishek	62000
Shubham	58000
Priya	67000
Rohit	64000
Neha	72000
Arman	55000
Ravi	60000
Sneha	75000
Kiran	70000
Tanuja	65000